

Exploratory Data Analysis Report

Swiggy Food Delivery Dataset · Delivery Time Prediction

Prepared by: Data Analyst | Dataset: Swiggy.csv | Analysis Scope: Full Column-Wise EDA

1. Introduction

This report presents the findings of a structured Exploratory Data Analysis (EDA) conducted on a food delivery dataset sourced from Swiggy, one of India's leading on-demand food delivery platforms. The central objective of the analysis is to understand the factors that influence delivery time, which serves as the target variable throughout the study.

The analysis follows a systematic column-wise examination that spans univariate, bivariate, and multivariate dimensions. Each variable is assessed individually for its distributional properties and then explored in relation to delivery time using appropriate statistical tests — including one-way ANOVA, independent samples t-tests, Pearson and Spearman correlations, chi-square tests of association, and simple linear regression. The EDA is implemented through a reusable function template that standardises the visual and statistical treatment applied across all variables.

The sections that follow present findings thematically, grouping variables by the nature of the insight they provide: target variable characteristics, operational factors, environmental conditions, rider and vehicle attributes, temporal patterns, and cross-variable interactions.

2. Target Variable — Delivery Time

The target variable, delivery time, forms the nucleus of the entire analysis. Its distributional properties are examined first to establish the baseline against which all other variables are evaluated.

Mean Delivery Time ~26 min Approximate average	Median Delivery Time ~26 min Consistent with mean	Skewness 0.48 Near-symmetric distribution	Distribution Shape Bimodal Peaks at 17–18 & 26–27 min
---	--	--	--

The delivery time variable exhibits a bimodal distribution with two discernible peaks — the first at approximately 17–18 minutes and the second at 26–27 minutes. Despite this bimodality, the mean and median values converge at approximately 26 minutes, indicating that the central tendency is stable. The skewness value of 0.48 falls within the acceptable range for near-normality, and the kurtosis values are also low. Consequently, the outliers observed in the upper tail are considered manageable and are not treated as a critical concern for downstream modelling.

3. Operational Factors

Operational factors are those directly tied to the logistics of a delivery — where the order is going, how far it must travel, and how many other deliveries are bundled together.

3.1 Delivery Distance (distance_km)

Distance in kilometres is among the most intuitively compelling predictors of delivery time. The distribution of delivery distances is multimodal, with values behaving almost categorically rather than continuously. Statistical testing confirms that distance has a significant impact on delivery time.

Mean Distance	9 km
Standard Deviation	5.5 km
Distribution Character	Multimodal — quasi-categorical spacing
Statistical Significance	Confirmed significant impact on delivery time
Distance Categories	Short and medium distances yield lower median delivery times

When distance is treated as a categorical variable (short, medium, long), a clear ordering emerges: short and medium distances are associated with lower median delivery times, while longer distances predictably extend the time required.

3.2 Multiple Deliveries

The multiple deliveries variable captures how many simultaneous orders a rider is fulfilling. The analysis reveals a strong monotonic relationship — delivery time increases consistently as the number of concurrent deliveries increases.

- Single deliveries (1 concurrent order) represent the highest proportion of observations and have the lowest associated delivery times.
- Orders with 2 or 3 simultaneous deliveries span a time range of 40–50 minutes, substantially higher than single-delivery orders.
- The sample sizes for 2 and 3 concurrent deliveries are limited, leading to higher variance. Merging these two categories is considered appropriate to improve statistical stability.

3.3 City Type

The dataset is heavily skewed toward metropolitan cities, which account for 76% of all deliveries. The median delivery time for metropolitan areas is 26–27 minutes. Semi-urban areas represent a very small fraction of the data (0.4%), making it difficult to draw reliable conclusions for that category. Urban areas constitute the remaining portion of observations.

4. Environmental Factors

Environmental conditions — weather, traffic, and festival periods — represent external forces beyond operational control that nonetheless exert measurable influence on how long deliveries take.

4.1 Traffic Conditions

Traffic is identified as one of the most significant predictors of delivery time. The mean delivery time varies clearly across traffic levels, making this variable an important feature for prediction.

No Traffic	~20 minutes median delivery time
Low Traffic	Intermediate delivery time
Medium Traffic	Intermediate delivery time
Heavy Traffic / Jam	~30 minutes median delivery time

The 10-minute gap between the no-traffic and traffic-jam conditions underscores the dominance of road conditions in determining how quickly an order reaches its destination.

4.2 Weather Conditions

Delivery time is broadly distributed across all weather categories, suggesting that weather alone does not deterministically extend or compress delivery times. However, ANOVA testing confirms that weather conditions as a group have a statistically significant impact on delivery time.

- Sunny weather is associated with the lowest median delivery time of approximately 20 minutes, and the effect holds even when traffic is controlled for.
- Stormy, sandstorm, and windy conditions exhibit similar distributional patterns and can be grouped together as adverse weather conditions.
- High-value outliers in delivery time are most prominent under sunny, stormy, and sandstorm conditions.

4.3 Festival Periods

Festival periods produce the most dramatic shift in delivery time of any single categorical variable. However, the volume of festival-day observations is extremely limited at approximately 2% of the dataset, which tempers the reliability of this finding.

- Median delivery time during non-festival periods: approximately 25 minutes.
- Median delivery time during festival periods: approximately 45 minutes — an increase of 80%.
- Festival days also record a higher prevalence of traffic jams, compounding the time effect.

5. Rider Characteristics

Rider-level attributes — age and performance rating — offer insight into how individual rider characteristics relate to delivery efficiency.

5.1 Rider Age

The age distribution of delivery riders is uniform, indicating no strong concentration in any particular age group. Despite the visually flat relationship between age and delivery time, statistical testing reveals a significant association between the two variables.

- Rider age is uniformly distributed, showing no dominant age cohort in the dataset.
- The line chart of delivery time against age displays a steep piecewise upward break in the 28–30 year age range, suggesting a nonlinear age effect in that interval.
- Statistical tests confirm some relationship between age and delivery time, even though the visual pattern does not show a simple or obvious linear trend.

5.2 Rider Ratings

The ratings variable presents distributional challenges. It is left-skewed with multimodality, meaning that ratings cluster in certain bands rather than spreading smoothly. Outlier values are also present at the lower end of the rating scale.

- A square transformation was applied to ratings in an attempt to improve distributional normality, but this did not produce a meaningfully better shape.
- Despite its distributional irregularities, statistical testing confirms that ratings have a significant impact on delivery time — riders with different rating levels do not deliver in the same time on average.

6. Vehicle and Order Attributes

The physical means of delivery — the type and condition of the vehicle — and the nature of the order itself are examined here for their impact on delivery time.

6.1 Vehicle Type

Vehicle type is a statistically significant predictor of delivery time. The dataset is dominated by motorcycles, which account for 58% of all deliveries.

Motorcycle	58% of deliveries; median delivery time 27–28 minutes; highest median time and most outliers
Electric Scooter	Lowest mean delivery time across all traffic conditions; fewest outliers
Other Types	Intermediate delivery time distributions

The electric scooter's superior delivery time performance holds regardless of traffic conditions, suggesting an inherent efficiency advantage for this vehicle type in the delivery context.

6.2 Vehicle Condition

Vehicle condition is coded numerically (0–3), with higher values presumably representing better condition. The analysis reveals a counterintuitive pattern: vehicles rated at condition 3 record the

highest median delivery time at 27–28 minutes, while all other condition ratings display broadly similar distributions. Vehicle condition 3 maintains this elevated delivery time regardless of the traffic situation.

6.3 Order Type

Order type is the only categorical variable in the study that demonstrates no statistically significant relationship with delivery time. Each order type category displays a similar distribution and nearly identical central tendency. This variable is therefore considered a weak predictor for delivery time modelling.

7. Temporal Patterns

A set of temporal variables captures when deliveries are made — including the time of day, day of the week, day of the month, month, and whether the delivery falls on a weekend. Pickup time is also examined. These variables are evaluated for patterns in order volume and their relationship with delivery time.

7.1 Time of Day

Time of day is one of the most behaviorally meaningful temporal variables in the dataset, connecting natural demand cycles with delivery performance.

Morning	Lowest median delivery time: 22–23 minutes; lowest traffic
Afternoon	Intermediate delivery times
Evening	Highest order volume; highest median delivery time: 30 minutes
Night	High traffic and delivery times comparable to evening

Evening and night hours combine high order volume with elevated traffic, producing the longest typical delivery windows. Morning deliveries benefit from minimal traffic, resulting in the shortest times.

7.2 Day of the Week

- Thursday records the highest number of orders across the week.
- Tuesday records the lowest order volume and also the lowest median delivery time at approximately 24 minutes.
- Friday records the highest median delivery time at 27–28 minutes.

7.3 Weekend vs. Weekday

The difference in delivery time between weekend and weekday orders is minimal. While the median delivery time is marginally lower on weekends, the mean delivery time is the same for both, and the distributional shapes are statistically similar. Weekend status does not appear to be a meaningful predictor of delivery time on its own.

7.4 Day of Month and Month

Two temporal variables — day of month and calendar month — show concentrations in order volume but do not reveal meaningful variation in delivery time:

- The 3rd and 4th days of the month account for approximately 42% of total orders.

- February and March together account for approximately 62% of total orders.
- Neither variable provides sufficient explanatory information about delivery time variation, and both are considered candidates for removal from the predictive modelling feature set.

7.5 Order Pickup Time

Order pickup time — the time elapsed between order placement and rider pickup — shows neither a visual nor a statistically significant relationship with delivery time. This variable is effectively an uninformative predictor in the current analysis.

8. Multivariate Interactions

The multivariate analysis explores how pairs of categorical variables interact in their combined effect on delivery time. This stage reveals compound effects that would be missed by examining variables individually.

8.1 Weather and Traffic

The interaction between weather and traffic produces some of the most extreme delivery times observed in the dataset:

- Cloudy or foggy conditions combined with heavy traffic produce the highest mean delivery times, reaching approximately 36 minutes.
- Sunny weather consistently delivers the lowest delivery times regardless of the traffic condition prevailing at the time.
- The lowest mean delivery time of 20 minutes is recorded under sunny weather with medium traffic.

8.2 Vehicle Type and Traffic

The vehicle type advantage observed in the univariate analysis persists across all traffic conditions in the multivariate setting:

- Electric scooters maintain the lowest mean delivery time across all traffic levels and also generate fewer extreme outlier observations.
- Motorcycles — the most prevalent vehicle type — record the highest median delivery times and the widest spread of outliers across all traffic conditions.

8.3 Vehicle Condition and Traffic

Vehicle condition 3 maintains its elevated delivery time across all traffic situations, confirming that its higher delivery time is not a traffic-confounded artefact but a consistent feature of that condition category.

8.4 Festival Periods and Traffic

Festival days amplify the effect of traffic on delivery time. Regardless of the traffic level, delivery times are uniformly higher on festival days. Additionally, traffic jam conditions are more prevalent during festivals than on non-festival days, suggesting that festival events both directly increase delivery time and indirectly do so through worsening road conditions.

8.5 City Type and Traffic

The distribution of median delivery time is broadly similar across metropolitan, urban, and semi-urban settings, irrespective of the traffic condition. Semi-urban areas show a slightly higher mean delivery time, but the limited data available for semi-urban orders constrains the confidence that can be placed in this observation.

8.6 Time of Day and Traffic

The interaction between time of day and traffic reveals a clear temporal gradient in both congestion and delivery speed:

- Morning hours experience the least traffic, and when there is no traffic in the morning, the mean delivery time falls to approximately 19 minutes — the lowest recorded in the dataset.
- Evening and night hours experience the heaviest traffic, and under jam conditions during these periods, the mean delivery time reaches 31 minutes.

9. Summary of Key Findings

The following synthesises the most important takeaways from the full EDA across all variable categories.

Key Findings at a Glance
✓ Delivery time averages ~26 minutes with a bimodal distribution and a low skewness of 0.48.
✓ Traffic conditions are a primary driver: delivery time ranges from ~20 min (no traffic) to ~30 min (jam).
✓ Festival days cause the sharpest single spike in delivery time — from ~25 to ~45 minutes.
✓ Distance is a statistically significant predictor; mean delivery distance is 9 km (SD: 5.5 km).
✓ Multiple deliveries progressively increase delivery time; bundles of 2–3 take 40–50 minutes.
✓ Electric scooters deliver fastest regardless of traffic; motorcycles (58% of fleet) record the highest times.
✓ Vehicle condition 3 exhibits the highest median delivery time across all traffic scenarios.
✓ Evening is the peak demand period and also records the highest median delivery time of 30 minutes.
✓ Morning deliveries are fastest, with a minimum mean of ~19 minutes under no-traffic conditions.
✓ Order type has no impact on delivery time; day of month, month, and pickup time are weak predictors.
✓ Rider ratings show statistical significance despite distributional irregularities.
✓ Cloudy/foggy weather combined with heavy traffic yields extreme delivery times (~36 minutes mean).

10. Feature Relevance Overview

This section consolidates the findings from two distinct analytical perspectives: (a) the descriptive statistics and statistical significance tests conducted during the EDA phase, and (b) the feature importance scores generated by three supervised machine learning models — Random Forest (RF), Gradient Boosting Machine (GBM), and Recursive Feature Elimination with Random Forest (RFE). The averaged importance scores across all three models form the final combined ranking. Where the two perspectives agree, they reinforce confidence in the feature's relevance. Where they diverge, the discrepancy reveals important nuances about each variable's role.

10.1 Model Feature Importance — Combined Ranking

Three models were fitted on 80% of the cleaned dataset, with 20% held out for evaluation. Numerical features (age, ratings, distance_km) were standardised before training. Categorical variables were label-encoded. Feature importance scores from all three models were merged and averaged to produce a single combined ranking. The results partition features into two bands:

- Top-5 features — ratings, weather, distance_km, multiple_deliveries, age — account for approximately 70% of combined feature importance.
- Adding traffic, vehicle_condition, time_of_day, and festival brings cumulative importance to approximately 95% of the total.
- The remaining six features — day_of_week, order_type, city_type, pickup_time_cat, vehicle_type, and is_weekend — contribute the bottom 5% and were excluded from a sub-feature model. That model posted an MAE of 3.15 minutes and R^2 of 81.9%, compared to 3.05 minutes and 83.0% with all features, confirming these six variables carry minimal predictive value.

10.2 EDA vs. Model Feature Importance — Side-by-Side Comparison

The table below places the EDA-derived relevance judgement alongside the model's combined importance tier for each variable, and flags whether the two approaches agree or diverge.

Variable	EDA Relevance	Model Tier	Agreement / Notes
Ratings	MODERATE	TOP-5 (~70%)	DIVERGE — Models rank ratings #1; EDA flagged distributional issues (left-skew, multimodality) but statistical tests also showed significance.
Weather	MODERATE	TOP-5 (~70%)	DIVERGE — EDA noted even distribution across weather types; models recognise weather as a strong nonlinear predictor.
Distance (km)	HIGH	TOP-5 (~70%)	AGREE — Both perspectives confirm distance as a key driver of delivery time.
Multiple Deliveries	HIGH	TOP-5 (~70%)	AGREE — Clear monotonic increase in EDA is confirmed by high model importance.
Rider Age	LOW-MODERATE	TOP-5 (~70%)	DIVERGE — EDA showed a flat visual pattern; models elevate age to top-5, likely capturing nonlinear effects invisible to univariate tests.

Traffic	HIGH	MID (~95%)	PARTIAL — EDA placed traffic as the primary driver; models rank it 6th. Its effects may be partially captured by time_of_day and distance.
Vehicle Condition	MODERATE	MID (~95%)	AGREE — Moderate importance in both; condition 3 anomaly noted in EDA aligns with mid-tier model ranking.
Time of Day	HIGH	MID (~95%)	PARTIAL — EDA highlighted a clear morning-to-evening gradient; models place it in the mid tier, possibly correlated with traffic.
Festival	HIGH	MID (~95%)	PARTIAL — The 80% spike in EDA is compelling but only 2% of observations are festival days; sparse data limits model weight.
Vehicle Type	HIGH	LOW (Dropped)	DIVERGE — EDA found electric scooters meaningfully faster; models rank it low, possibly because motorcycle dominance (58%) limits variance.
Day of Week	LOW	LOW (Dropped)	AGREE — Both agree this variable carries minimal predictive signal.
Is Weekend	LOW	LOW (Dropped)	AGREE — Marginal difference in EDA; dropped by models with negligible accuracy cost.
City Type	LOW	LOW (Dropped)	AGREE — Metropolitan dominance (76%) limits within-class variation; both agree it is not useful.
Order Type	NOT SIGNIFICANT	LOW (Dropped)	AGREE — Consistently the weakest variable across both statistical tests and model scores.
Pickup Time (cat)	NOT SIGNIFICANT	LOW (Dropped)	AGREE — No visual or statistical impact in EDA; model confirms negligible importance.

AGREE — EDA and model importance concur

PARTIAL — Broad agreement, but model ranking differs from EDA priority

DIVERGE — EDA and model importance point in opposite directions

10.3 Key Divergences and Interpretations

Several variables produced meaningfully different signals between the two analytical approaches. The following observations are drawn directly from the feature selection notebook:

- Rider Ratings emerged as the single most important feature in the combined model ranking, a finding that contrasts with the EDA classification of MODERATE relevance. The EDA noted left-skew, multimodality, and the failure of a square transformation to normalise the variable — yet statistical tests during the EDA already confirmed a significant impact on delivery time. The model result suggests that ratings encode predictive signal that is richer than its distributional shape

implies.

- Rider Age, assigned LOW-MODERATE relevance in the EDA due to its uniform distribution and modest linear relationship, appears among the top-5 features by combined model importance. The EDA did note a piecewise nonlinear break in the 28–30 age range; tree-based models are well-positioned to exploit such discontinuities, which likely explains the discrepancy.
- Weather, treated as MODERATE in the EDA because delivery time appeared evenly distributed across conditions, is ranked second by the models. This suggests that weather's effect manifests as interaction effects and threshold patterns that tree ensembles detect but simple group-mean comparisons understate.
- Traffic, which the EDA identified as the primary operational driver with a 10-minute spread from no-traffic to jam conditions, falls to the middle tier in the model ranking. This may reflect multicollinearity between traffic and time_of_day — both capture congestion-related variance — causing the model to distribute importance across the two rather than concentrating it in traffic alone.
- Vehicle Type was rated HIGH in the EDA, with electric scooters consistently recording the lowest delivery times. However, the models dropped vehicle_type from the sub-feature set. The motorcycle's 58% dominance of the fleet severely compresses the within-category variance available for learning, likely suppressing its apparent importance in tree-based feature scoring.

This report documents the complete findings of the Exploratory Data Analysis and the model-based feature selection conducted on the Swiggy delivery dataset. All conclusions are derived directly from the statistical outputs and visual analyses produced in the EDA and Feature Selection notebooks. Together, the two analytical lenses provide a complementary and more robust foundation for feature engineering and predictive modelling of food delivery times.